STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 5: Work, Energy and Power

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. When is work zero with displacement?
2. Kinetic energy?
3. SI unit of power?
4. Work-energy theorem?
5. Spring energy?
6. What is conserved in an isolated elastic collision?
7. Near-earth gravitational potential energy?
8. Work by centripetal force in uniform circular motion?

Section B 2-mark questions - 6 x 2 = 12 marks

9. N over 4 m along force: work?
10. kg at 4 m/s: K?
11. W for 20 s: energy?
12. N perpendicular to 5 m displacement: work?
13. k=200 N/m, spring stretched .1 m: energy?
14. kg cart at 5 m/s sticks to stationary 4 kg: speed?

Section C 3-mark questions - 4 x 3 = 12 marks

15. kg speeds from 2 to 6 m/s: net work?
16. kg falls 3 m, g=10: final speed?
17. F=4x N from x=0 to 3 m: work?
18. kg car at 20 m/s stops in 50 m: W and force magnitude?

Section D 4-mark questions - 2 x 4 = 8 marks

19. kg drops 5 m frictionlessly, g=10: speed; with 20 J loss?
20. kg cart at 8 m/s sticks to 3 kg stationary: speed and energy loss?